

Interactively Visualizing Multivariate Market Segmentation Using the R Package Lionfish: Response to Review

We sincerely thank the editor and reviewers for your valuable suggestions, which have greatly improved the quality of both the article and the lionfish software.

General remark

The following summarises all of our changes and explanations to the comments provided. *Reviewer's comments are shown in italic.* When a comment resulted in specific changes in the text, we have added the original as well as the revised text to the response to that comment. Some additional changes were made to the paper, based on our own re-reading of the paper.

Reviewer A

- *I'm satisfied with the changes - nice work!*

Response: Thank you

Reviewer B

General Comments

- *In this version, the goals of the paper are much clearer. And, I was able to run the software, with just some minor issues. Software documentation could do with*

further improvement. Overall, this revision is significantly improved.

I do not see the need to include both the Austrian and Australian tourism examples in the paper, as they illustrate practically the same features of the software. While online journals may not be as strict about page limits, repetitive content could be annoying for readers.

Some minor points of clarification are listed below. Further very minor suggestions are on the paper pdf.

Response:

- *In the beginning of section 2, explain what is meant by grand tour and manual tour.*

Response: Done

- *In section 2 it states that python was chosen "for high-performance interactivity". Could shiny not have been used for this purpose?*

Response: Done

- *The GUI for the tour via menus E, F and H on Figure 4 is not at all clear.*

Response:

- *Justify the choice of kmeans for binary data and likert data.*

Response:

- *Give a rationale for the choice of kmeans $k=5/6$ in your examples.*

Response:

- *In Figure 13 a and b, the axes for Friends are in opposite directions, please explain.*

Response:

- *Cluster 1 has disappeared from figure 14, worth a comment.*

Response:

- *The so-called "gold" colour does not look a bit gold... more chartreuse???*

Response:

- *Figure 15: it is not true that "all axes generally point in the same direction"*

Response:

Software

- *the documentation example for interactive_ tour fails (f not defined)*

Response:

- *After running `init_ env()` (Connection to python backend established), interactive_ tour opens a window with title "interactive tour" and with blank content*

Response:

- *Using `init_ env( virtual_ env = "anaconda")` I can run some of the examples as described in the paper.*

Response:

- *The example given here <https://mmedl94.github.io/lionfish/> needs more explanation.*

Response:

- *hitting stop button in Console crashes R*

Response:

- *In the projection arrows for the histogram, a marker line for "zero" would be helpful.*

Response:

- *How about colouring the cluster labels in the heatmaps?*

Response:

- *The code for generating the results in the "saves" folder is not provided.*

Response: